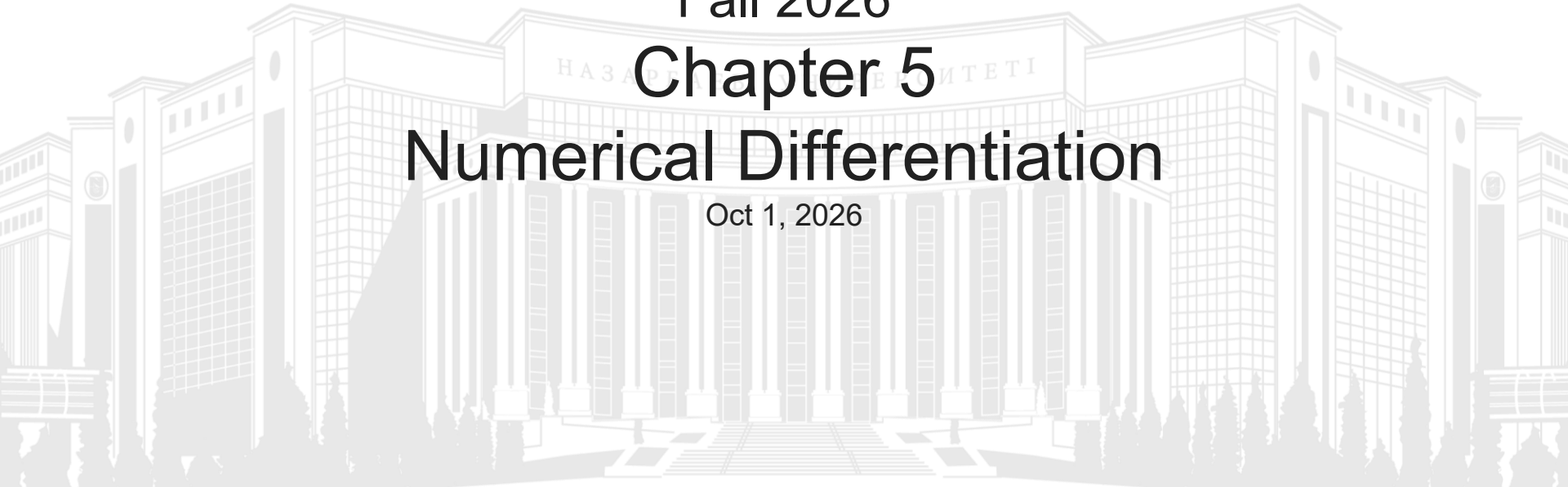


NAZARBAYEV UNIVERSITY

Numerical Methods with Applications
Math 351 – 1L

Fall 2026

Chapter 5
Numerical Differentiation

Oct 1, 2026

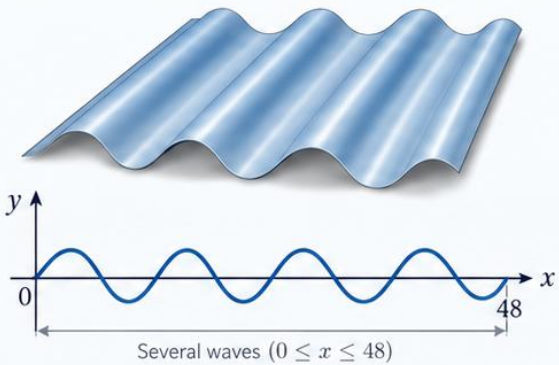
Motivation: Why Numerical Integration?

Numerical integration and differentiation are fundamental tools in scientific computing.

1 Problem setup / model

Example: estimate the length of a corrugated roofing sheet whose profile is approximated by a sine wave.

$$f(x) = \sin(x), \quad x \in [0, 48].$$



2 Arc-length formula from calculus

From calculus, the length of a curve $y = f(x)$ on $[a, b]$ is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx.$$

For our model $f(x) = \sin(x)$, the arc length is

$$L = \int_0^{48} \sqrt{1 + [f'(x)]^2}\, dx$$
$$= \int_0^{48} \sqrt{1 + \cos^2(x)}\, dx.$$

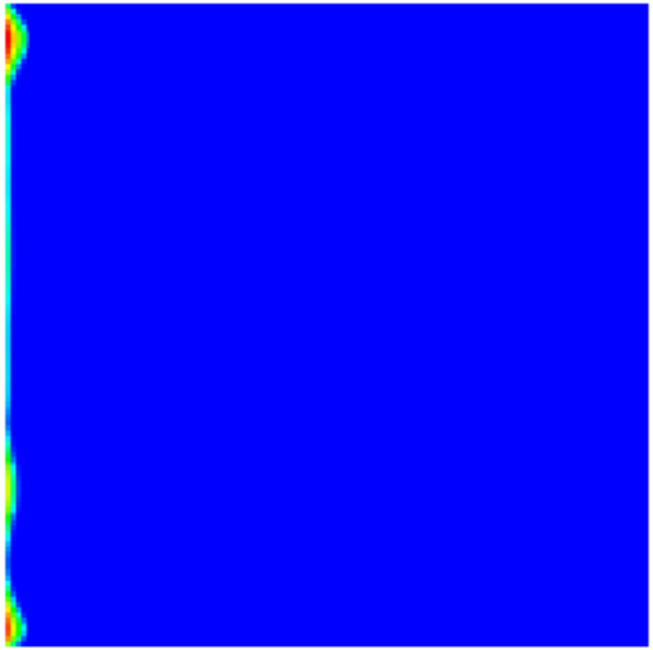
3 Why numerical integration is needed

- The integral $\int_0^{48} \sqrt{1 + \cos^2(x)}\, dx$ does not have an elementary antiderivative.
- It cannot be expressed in terms of standard elementary functions (e.g., polynomials, exponentials, trigonometric functions, etc.).

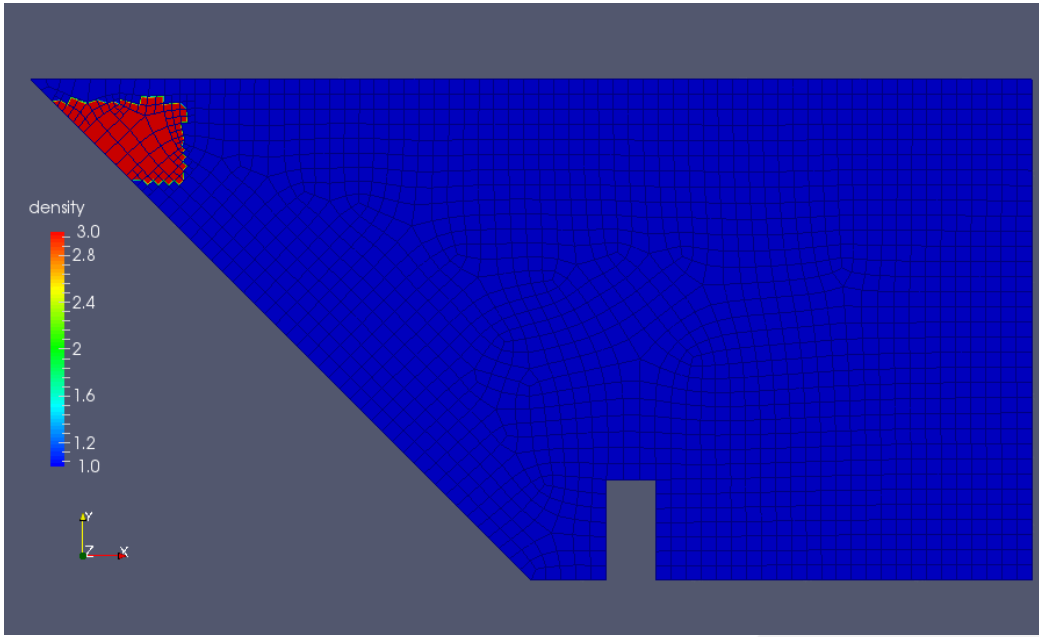
No elementary antiderivative

The integral must be approximated numerically.

Therefore, we approximate L numerically using quadrature methods (e.g., Trapezoidal Rule or Simpson's Rule).



Two phase flow in porous media, e.g. Oil and Water

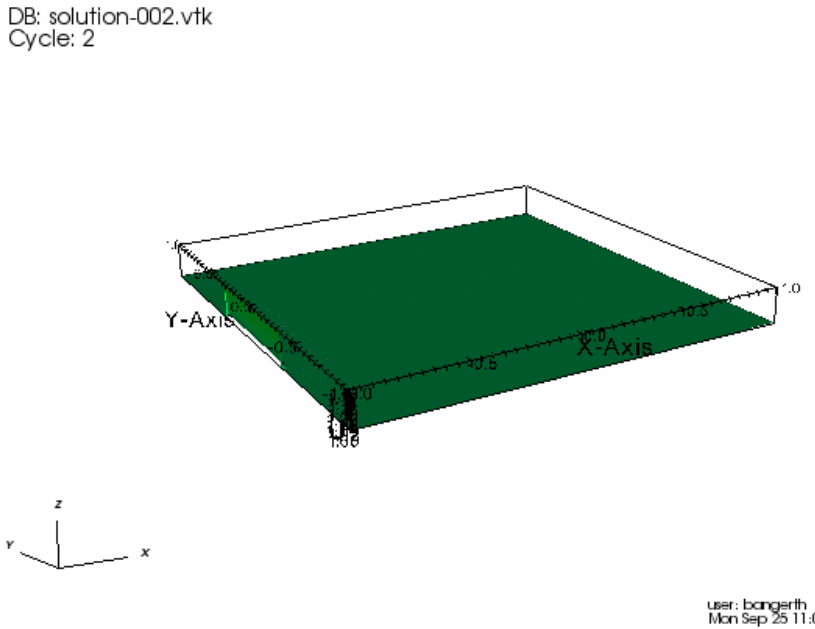


Gas flow model

Partial differential equations

- Before we concentrate on methods for initial value ODEs, here are some PDE prototypes for a more complete general picture.
- Simplest elliptic PDE: Poisson.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = g(x, y).$$



- Simplest parabolic PDE: heat.

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}.$$

- Simplest hyperbolic PDE: wave.

$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0.$$

Motivation: Why Numerical Differentiation?

1 Goal

Develop numerical approximations of derivatives such as $f'(x)$, $f''(x)$, etc., using only function values of $f(x)$.

2 Derivative from calculus

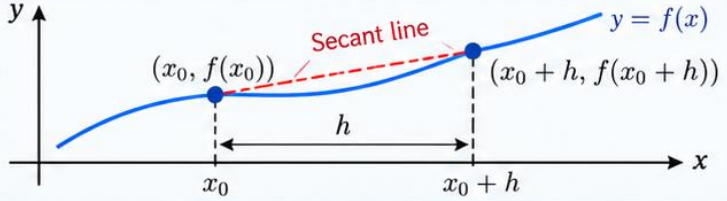
Recall that the derivative of $f(x)$ at x_0 is defined by

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

3 Numerical approximation idea

If h is small, we may replace the limit by a finite difference:

$$f'(x_0) \approx \frac{f(x_0 + h) - f(x_0)}{h}.$$



4 Key question

How small should h be?

Choosing h too large increases truncation error, while choosing h too small may increase round-off error.

Numerical differentiation balances accuracy and stability.

This motivates the study of forward, backward, and central difference formulas.

- We use the formula above to approximate the derivative of $f(x) = \sin(x)$ at $x = 1$. Note that the true value is $f'(1) = \cos(1)$.
- We test with $h = \frac{1}{10}, \frac{1}{100}, \cdots, \frac{1}{10^{15}}$, and find the relative error.

A MATLAB Script

```
% Test the derivative approximation
% f(x) = sin(x), f'(x) = cos(x).
% This example we will approximation f'(1) = cos(1)
clear
format long
h = zeros(15,1);
err = zeros(15,1);
for i = 1:15
    h(i) = 10^(-i);
    app = (sin(1+h(i))-sin(1))/h(i);
    err(i) = abs(app - cos(1))/cos(1);
end
disp('   h                    relative error')
disp([h,err])
```

Numerical Result

```
>> ex4_1_0
   h                    relative error
   0.100000000000000    0.079471349402736
   0.010000000000000    0.007803640314835
   0.001000000000000    0.000778870464261
   0.000100000000000    0.000077872052482
   0.000010000000000    0.000007787052229
   0.000001000000000    0.000000778724808
   0.000000100000000    0.000000077415348
   0.000000010000000    0.000000005496710
   0.000000001000000    0.000000097244201
   0.000000000100000    0.000000108237621
   0.000000000010000    0.000002163055846
   0.000000000001000    0.000080029673119
   0.000000000000100    0.001358343083757
   0.000000000000010    0.006860929812673
   0.000000000000001    0.027409112053748
```

- At the beginning, the relative error gets smaller as h decreases.
- After the eighth iteration, $h = 10^{-8}$, the error starts to grows as we further reduce the size of h. **Why?**

Derivation of the Forward Difference Formula

Idea: Approximate $f(x)$ locally by a linear interpolating polynomial and differentiate it.

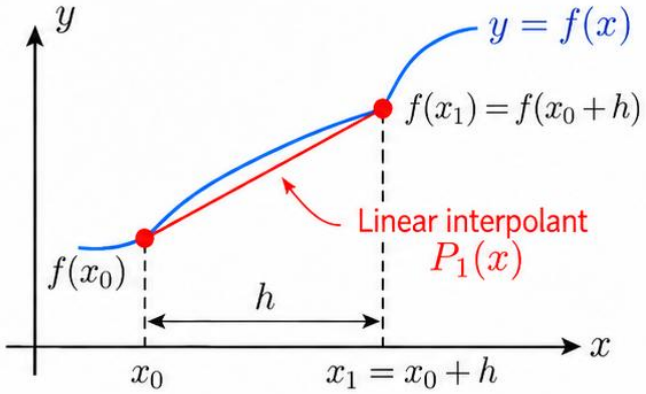
1 Linear interpolation at x_0 and $x_1 = x_0 + h$

The linear Lagrange basis polynomials are

$$L_{1,0}(x) = \frac{x - x_1}{x_0 - x_1}, \qquad L_{1,1}(x) = \frac{x - x_0}{x_1 - x_0}.$$

With $x_1 = x_0 + h$, these become

$$L_{1,0}(x) = \frac{x - (x_0 + h)}{-h} = \frac{x_0 + h - x}{h}, \qquad L_{1,1}(x) = \frac{x - x_0}{h}.$$



2 Linear Lagrange interpolation polynomial

The linear interpolating polynomial $P_1(x)$ of $f(x)$ is

$$\begin{aligned} P_1(x) &= f(x_0)L_{1,0}(x) + f(x_1)L_{1,1}(x) \\ &= f(x_0)\frac{x_0 + h - x}{h} + f(x_0 + h)\frac{x - x_0}{h} \\ &= f(x_0) + \frac{f(x_0 + h) - f(x_0)}{h}(x - x_0). \end{aligned}$$

$P_1(x)$ is the **straight line** through the two points on the curve $y = f(x)$:

$(x_0, f(x_0))$ and $(x_0 + h, f(x_0 + h))$.

3 Differentiate the interpolating polynomial

Taking the derivative, we get

$$P_1'(x) = \frac{f(x_0 + h) - f(x_0)}{h}.$$

Hence, **we approximate**

$$f'(x_0) \approx P_1'(x_0) = \frac{f(x_0 + h) - f(x_0)}{h},$$

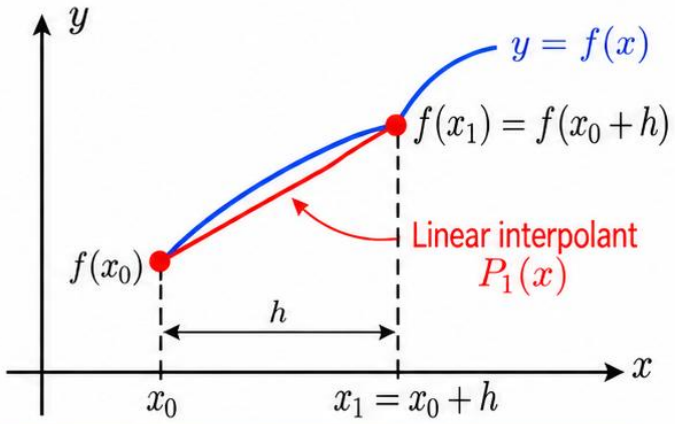
(forward difference formula).

Error Bound for the Derivative (Forward Difference)

- Recall the error bound of linear Lagrange interpolation (ref. Sec 3.1) is

$$f(x) - P_1(x) = \frac{f''(\xi(x))}{2!}(x - x_0)(x - x_1),$$

where $\xi(x)$ is between x_0 and x_1.



- So, the error bound of the derivative is

$$\begin{aligned} f'(x) - P_1'(x) &= D_x\left(\frac{f''(\xi(x))}{2!}(x - x_0)(x - x_1)\right). \\ &= \frac{1}{2}(2x - x_0 - x_1)f''(\xi) + \frac{(x - x_0)(x - x_1)}{2} D_x(f''(\xi)) \end{aligned}$$

where ξ is between x_0 and x.

- When $x = x_0$, the second term on the right hand side is zero. We have

$$f'(x_0) - P_1'(x_0) = -\frac{h}{2}f''(\xi),$$ where ξ is between x_0 and x_1 $(= x_0 + h)$.

Forward/Backward Difference Formula

$$f'(x_0) \approx \frac{f(x_0+h)-f(x_0)}{h}. \tag{4.1}$$

- If $h > 0$, it is called **forward-difference formula**.
- If $h < 0$, it is called **backward-difference formula**.

Error Bound for Forward/Backward Difference Formula

For small value of h, we have

$$\left| f'(x_0) - \frac{f(x_0+h)-f(x_0)}{h} \right| \leq \frac{M|h|}{2}, \tag{4.2}$$

where M is the bound for $|f''(x)|$ for x between x_0 and x_0+h. We say that the **truncation error** is $\mathcal{O}(h)$.

Numerical Differentiation: Example (Forward Difference)

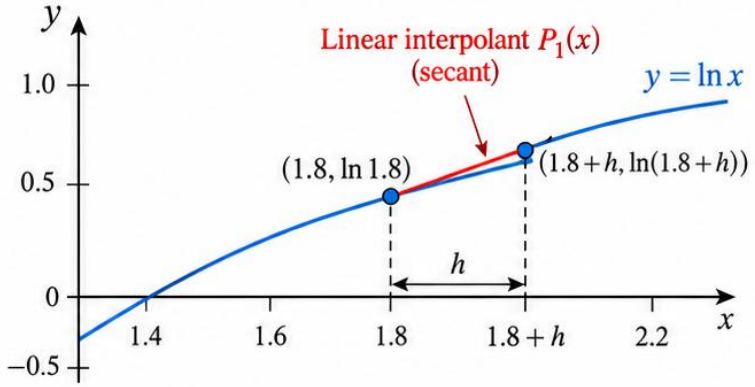
Example: Approximate $f'(1.8)$ using the forward difference formula for $f(x) = \ln x$.

$$f'(x_0) \approx \frac{f(x_0+h) - f(x_0)}{h}, \qquad x_0 = 1.8.$$

Exact derivative: $f'(x) = \dfrac{1}{x}$,

$$f'(1.8) = \frac{1}{1.8} = 0.5555556.$$

h	$f(1.8+h)$	$\dfrac{f(1.8+h) - f(1.8)}{h}$	$\dfrac{\lvert h\rvert}{2(1.8)^2}$
0.1	0.64185389	0.5406722	0.0154321
0.05	0.61518564	0.5479795	0.0077160
0.01	0.59332685	0.5540180	0.0015432



Summary:

- As $h \to 0$, the forward difference approximation approaches the true value $f'(1.8) = 0.5555556$.
- The error is first-order: $O(h)$. The last column gives the bound $\dfrac{\lvert h\rvert}{2(1.8)^2}$.

General $(n+1)$-Point Formula for Numerical Differentiation

1 Lagrange interpolation polynomial

Let $P_n(x)$ be the Lagrange interpolation polynomial of $f(x)$ at $n+1$ distinct points $x_0, x_1, \ldots, x_n$. Then

$$P_n(x) = \sum_{k=0}^{n} f(x_k) L_{n,k}(x).$$

where

$$L_{n,k}(x) = \prod_{\substack{j=0 \\ j \neq k}}^{n} \frac{x - x_j}{x_k - x_j}.$$

2 Approximation of the derivative

The derivative of $f(x)$ can be approximated by

$$f'(x) \approx P_n'(x) = \sum_{k=0}^{n} f(x_k) L_{n,k}'(x). \qquad (4.3)$$

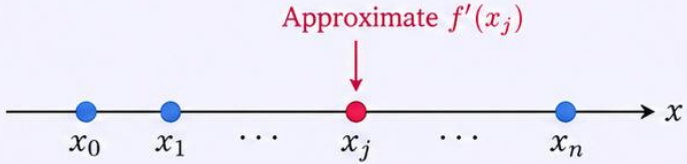
3 Error at a node

When $x = x_j$, the error is

$$f'(x_j) - P_n'(x_j) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{\substack{k=0 \\ k \neq j}}^{n} (x_j - x_k).$$

4 Interpretation

Equation (4.3) is called an $(n+1)$-**point formula** for approximating $f'(x_j)$.



- To approximate $f'(x)$, 3-point and 5-point formulas are commonly used.
- For **three-point formulas**, we have for each $j = 0, 1, 2$,

$$\begin{aligned} f'(x_j) &= P_2'(x_j) + \frac{f'''(\xi)}{3!} \prod_{\substack{j=0 \\ j \neq k}}^{2} (x_j - x_k). \\ &= f(x_0) \frac{2x_j - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)} + f(x_1) \frac{2x_j - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} \\ &\quad + f(x_2) \frac{2x_j - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)} + \frac{f'''(\xi)}{3!} \prod_{\substack{j=0 \\ j \neq k}}^{2} (x_j - x_k). \end{aligned}$$

- If the nodes are equally spaced, i.e., $x_1 = x_0 + h$, $x_2 = x_0 + 2h$, then

$$\begin{aligned} f'(x_0) &= -\frac{3}{2h} f(x_0) + \frac{1}{h} f(x_1) - \frac{1}{2h} f(x_2) + \frac{h^2}{3} f'''(\xi_0), \\ f'(x_1) &= -\frac{1}{2h} f(x_0) + \frac{1}{2h} f(x_2) - \frac{h^2}{6} f'''(\xi_1), \\ f'(x_2) &= \frac{1}{2h} f(x_0) - \frac{2}{h} f(x_1) - \frac{3}{2h} f(x_2) + \frac{h^2}{3} f'''(\xi_2). \end{aligned}$$

Three-Point Endpoint Formula for f'

$$f'(x_0) = \frac{1}{2h}\left[-3f(x_0) + 4(x_0 + h) - f(x_0 + 2h)\right] + \frac{h^2}{3}f'''(\xi) \tag{4.4}$$

where ξ lies between x_0 and $x_0 + 2h$.

Remark The other end-point formula can be obtained by replacing h with $-h$.

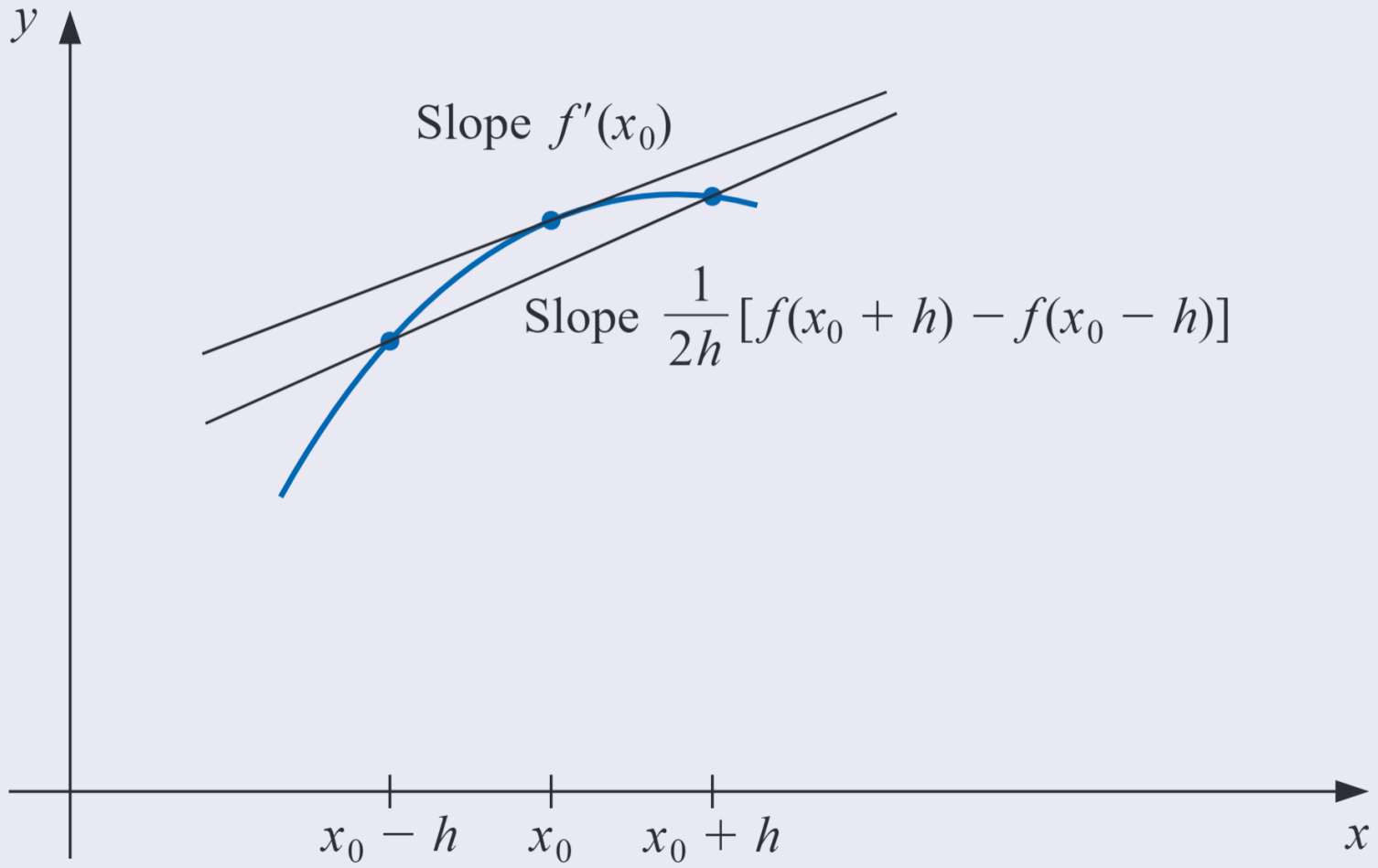
Three-Point Midpoint Formula for f'

$$f'(x_0) = \frac{1}{2h}\left[f(x_0 + h) - f(x_0 - h)\right] - \frac{h^2}{6}f'''(\xi) \tag{4.5}$$

where ξ lies between $x_0 - h$ and $x_0 + h$.

- **Remark** Although the truncation error for midpoint and endpoint formulas are both $\mathcal{O}(h^2)$, the error in (4.5) is approximately half the error in (4.4). This is because (4.5) uses data on both sides of x_0 while (4.4) only uses data from one side.

An illustration of the three-point midpoint approximation



Five-Point Midpoint Formula for f'

$$f'(x_0) = \frac{1}{12h}\Big[f(x_0-2h)-8(x_0-h)+8f(x_0+h)-f(x_0+2h)\Big]+\frac{h^4}{30}f^{(5)}(\xi). \quad (4.6)$$

where ξ lies between $x_0 - 2h$ and $x_0 + 2h$.

Five-Point Endpoint Formula for f'

$$f'(x_0) = \frac{1}{12h}\Big[-25f(x_0) + 48f(x_0+h) - 36(x_0+2h) + 16f(x_0+3h) - 3f(x_0+4h)\Big] + \frac{h^4}{5}f^{(5)}(\xi) \quad (4.7)$$

where ξ lies between x_0 and $x_0 + 4h$.

Remark

- The truncation errors of 5-point formulas (4.6) and (4.7) are $\mathcal{O}(h^4)$.
- The formula (4.7) is particularly useful for the clamped spline interpolation for approximating the derivative of the boundary points.

Given the following data for $f(x) = xe^x$.

x	1.8	1.9	2.0	2.1	2.2
$f(x)$	10.889365	12.703199	14.778112	17.148957	19.855030

Use three-point and five-point formulas to approximate $f'(2.0)$.

Solution (1/3)

- We first use 3-point endpoint formula (4.4) with $h = 0.1$,

$$\begin{aligned} f'(2.0) &\approx \frac{1}{0.2}\left[-3f(2.0) + 4(2.1) - f(2.2)\right] \\ &= 5[-3(14.778112) + 4(17.148957) - 19.855030] \\ &= 22.032310. \end{aligned}$$

Solution (2/3)

- We can also use 3-point endpoint formula (4.4) with $h = -0.1$,

$$\begin{aligned} f'(2.0) &\approx \frac{1}{-0.2}\left[-3f(2.0) + 4(1.9) - f(1.8)\right] \\ &= -5[-3(14.778112) + 4(12.703199) - 10.889365] \\ &= 22.054525. \end{aligned}$$

- Using 3-point midpoint formula (4.5) with $h = 0.1$ we have

$$f'(2.0) \approx \frac{1}{0.2}\left[f(2.1) - f(1.9)\right] = 5(17.148957 - 12.7703199) = 22.228790.$$

- Using 3-point midpoint formula (4.5) with $h = 0.2$ we have

$$f'(2.0) \approx \frac{1}{0.4}\left[f(2.2) - f(1.8)\right] = 2.5(19.855030 - 10.889365) = 22.41416.$$

Solution (3/3)

- The only 5-point formula applicable is the 5-point midpoint formula (4.6) with $h = 0.1$

$$\begin{aligned} f'(2.0) &\approx \frac{1}{1.2}\left[f(1.8) - 8f(1.9) + 8f(2.1) - f(2.2)\right] \\ &= \frac{1}{1.2}[10.889365 - 8(12.703199) + 8(17.148957) - 19.855030] \\ &= 22.166999. \end{aligned}$$

- The true derivative is $f'(2.0) = (2+1)e^2 = 22.167168$.
- The relative error of these approximations are
  - 3-pt endpoint with $h = 0.1$: 0.00608
  - 3-pt endpoint with $h = -0.1$: 0.00509
  - 3-pt midpoint with $h = 0.1$: 0.00277
  - 3-pt midpoint with $h = 0.2$: 0.01114
  - 5-pt midpoint with $h = 0.1$: 0.0000076

- Methods can also be derived to find approximations to higher derivatives of a function using only tabulated values of the function at various points.
- We will use Taylor expansion to derive the 2nd-order derivative approximation. Note that

$$\begin{aligned} f(x_0+h) &= f(x_0) + hf'(x_0) + \frac{h^2}{2}f''(x_0) + \frac{h^3}{6}f'''(x_0) + \frac{h^4}{24}f^{(4)}(\xi_1), \\ f(x_0-h) &= f(x_0) - hf'(x_0) + \frac{h^2}{2}f''(x_0) - \frac{h^3}{6}f'''(x_0) + \frac{h^4}{24}f^{(4)}(\xi_{-1}), \end{aligned}$$

  where $\xi_1 \in (x_0, x_0+h)$ and $\xi_{-1} \in (x_0-h, x_0)$.

- Adding these two equations we obtain

$$f(x_0+h) + f(x_0-h) = 2f(x_0) + h^2 f''(x_0) + \frac{h^4}{24}\Big(f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1})\Big).$$

- Subtracting $2f(x_0)$ on both side and dividing by h^2, we obtain

$$f''(x_0) = \frac{f(x_0+h) - 2f(x_0) + f(x_0-h)}{h^2} - \frac{h^2}{24}\Big(f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1})\Big).$$

- Using the Intermediate Value Theorem, if $f^{(4)}$ is continuous, then there exists $\xi \in (x_0 - h, x_0 + h)$ such that

$$f^{(4)}(\xi) = \frac{1}{2}\big(f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1})\big).$$

Midpoint Formula for f''

$$f''(x_0) = \frac{f(x_0+h) - 2f(x_0) + f(x_0-h)}{h^2} - \frac{h^2}{12}f^{(4)}(\xi). \tag{4.8}$$

for some $\xi \in (x_0 - h, x_0 + h)$.

Remark: The truncation error of the midpoint formula (4.8) is $\mathcal{O}(h^2)$.

Given the following data for $f(x) = xe^x$.

x	1.8	1.9	2.0	2.1	2.2
$f(x)$	10.889365	12.703199	14.778112	17.148957	19.855030

Use the second derivative midpoint formula to approximate $f''(2.0)$.

Solution

- Using the midpoint formula (4.8) with $h = 0.1$ we have

$$f''(2.0) \approx \frac{1}{0.1^2}\Big[f(1.9) - 2f(2.0) + f(2.1)\Big] = 29.593200.$$

- Using the midpoint formula with $h = 0.2$ we have

$$f''(2.0) \approx \frac{1}{0.2^2}\Big[f(1.8) - 2f(2.0) + f(2.2)\Big] = 29.704275.$$

- Because $f''(x) = (x + 2)e^x$, the exact value is $f''(2.0) = 29.556224$.
- The relative errors for $h = 0.1$ and 0.2 are 0.00125 and 0.00501, respectively.

Use three-point midpoint formula to approximate $f'(0.9)$ where $f(x) = \sin(x)$ with $h = 10^{-i}$, $i = 1, 2, \cdots, 12$.

Solution. (The screenshot of MATLAB command window)

Approximation of derivative of sin(x) at 0.9 using three-point midpoint

i	h	Approx	cos(0.9)	error
1	0.100000000000	0.620574469542	0.621609968271	0.001035498729
2	0.010000000000	0.621599608156	0.621609968271	0.000010360114
3	0.001000000000	0.621609864669	0.621609968271	0.000000103602
4	0.000100000000	0.621609967235	0.621609968271	0.000000001036
5	0.000010000000	0.621609968254	0.621609968271	0.000000000016
6	0.000001000000	0.621609968277	0.621609968271	-0.000000000006
7	0.000000100000	0.621609967943	0.621609968271	0.000000000327
8	0.000000010000	0.621609969054	0.621609968271	-0.000000000783
9	0.000000001000	0.621609985707	0.621609968271	-0.000000017436
10	0.000000000100	0.621609985707	0.621609968271	-0.000000017436
11	0.000000000010	0.621608320373	0.621609968271	0.000001647898
12	0.000000000001	0.621613871488	0.621609968271	-0.000003903217

The smallest error is around $i = 5, 6$, i.e., h is between 10^{-6} and 10^{-5}. **Why?**

- Let us examine the 3-pt midpoint formula:

$$f'(x_0) = \frac{f(x_0+h) - f(x_0-h)}{2h} - \frac{h^2}{6} f'''(\xi).$$

- When evaluating $f(x_0+h)$ and $f(x_0-h)$ with computer, there are round-off errors, denoted by $e(x_0+h)$ and $e(x_0-h)$. Then our computational actually use the values $\tilde{f}(x_0 \pm h)$ given by

$$f(x_0+h) = \tilde{f}(x_0+h) + e(x_0+h), \quad f(x_0-h) = \tilde{f}(x_0-h) + e(x_0-h).$$

- The total error in the approximation is

$$\begin{aligned} & f'(x_0) - \frac{\tilde{f}(x_0+h) - \tilde{f}(x_0-h)}{2h} \\ = \; & f'(x_0) - \frac{f(x_0+h) - f(x_0-h)}{2h} + \frac{e(x_0+h) - e(x_0-h)}{2h} \\ = \; & -\frac{h^2}{6} f'''(\xi) + \frac{e(x_0+h) - e(x_0-h)}{2h}. \end{aligned}$$

- The first term is the approximation error, and the second term is round-off error.

- If we assume that the round-off errors and the third derivative f''' are bounded as follows

$$|e(x_0 \pm h)| \leq \varepsilon, \qquad |f'''(x)| \leq M,$$

then the total computational error

$$\begin{aligned} & f'(x_0) - \frac{\tilde{f}(x_0+h) - \tilde{f}(x_0-h)}{2h} \\ \leq \quad & \frac{h^2}{6}|f'''(\xi)| + \frac{|e(x_0+h) - e(x_0-h)|}{2h} \\ \leq \quad & \frac{h^2}{6}|f'''(\xi)| + \frac{|e(x_0+h)| + |e(x_0-h)|}{2h} \\ \leq \quad & \frac{Mh^2}{6} + \frac{\varepsilon}{h}. \end{aligned}$$

- Note that as h is reduced, the approximation error $Mh^2/6$ decreases. However, the round-off error ε/h grows. If h is too small (e.g. $h < 10^{-6}$ as in previous example), the round-off error dominates the calculation.

- Now we analyze why h around 10^{-6} and 10^{-5} yields the most accurate approximation in Example 3. Note $f(x) = \sin(x)$, so $f'''(x) = -\cos(x)$.
- Around $x_0 = 0.9$, we consider the interval $[0.8, 1]$,

$$M = \max_{x\in[0.8,1]} |f'''(x)| = \max_{x\in[0.8,1]} |\cos(x)| = \cos(0.8) \approx 0.69671.$$

- In a 64-bit computer, the machine precision $\varepsilon \approx 2.22 \times 10^{-16}$. Hence, the total error

$$e(h) = \frac{Mh^2}{6} + \frac{\varepsilon}{h}$$

will reach its minimum where $e'(h) = \dfrac{Mh}{3} - \dfrac{\varepsilon}{h^2} = 0$. That is

$$h = \left(\frac{3\varepsilon}{M}\right)^{1/3} = \left(\frac{3 \times 2.22^{-16}}{0.69671}\right)^{1/3} \approx 0.0000098515.$$

- The value of h is **optimal** for the 3pt-midpoint formula on a 64-bit computer. This is consistent with the numerical experiment results, i.e., h is between 10^{-6} and 10^{-5}.

- Similar argument can be used to analyze why the forward difference formula reaches the best accuracy around $h = 10^{-8}$. (the first example of this section). (**Exercise**)

- In practice, we cannot compute an optimal h to use in approximating the derivative, since we have no knowledge of the derivatives of the function. But we must remain aware that reducing the step size h will not always improve the approximation.

- Numerical differentiation, as an approximation method, is **unstable**, since the small values of h needed to reduce truncation error also cause the round-off error to grow.

Three-Point Formulas

THREE-POINT ENDPOINT FORMULA

$$f'(x_0) = \frac{1}{2h}[-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)] + \frac{h^2}{3}f^{(3)}(\xi_0),$$

where ξ_0 lies between x_0 and $x_0 + 2h$.

THREE-POINT MIDPOINT FORMULA

$$f'(x_0) = \frac{1}{2h}[f(x_0 + h) - f(x_0 - h)] - \frac{h^2}{6}f^{(3)}(\xi_1),$$

where ξ_1 lies between $x_0 - h$ and $x_0 + h$.

Five-Point Formulas

FIVE-POINT MIDPOINT FORMULA

$$f'(x_0) = \frac{1}{12h}[f(x_0 - 2h) - 8f(x_0 - h) + 8f(x_0 + h) - f(x_0 + 2h)] + \frac{h^4}{30}f^{(5)}(\xi),$$

where ξ lies between $x_0 - 2h$ and $x_0 + 2h$.

FIVE-POINT ENDPOINT FORMULA

$$\begin{aligned} f'(x_0) &= \frac{1}{12h}[-25f(x_0) + 48f(x_0 + h) - 36f(x_0 + 2h) \\ &\quad + 16f(x_0 + 3h) - 3f(x_0 + 4h)] + \frac{h^4}{5}f^{(5)}(\xi), \end{aligned}$$

where ξ lies between x_0 and $x_0 + 4h$.

SECOND DERIVATIVE MIDPOINT FORMULA

$$f''(x_0) = \frac{1}{h^2}[f(x_0 - h) - 2f(x_0) + f(x_0 + h)] - \frac{h^2}{12}f^{(4)}(\xi),$$

for some ξ, where $x_0 - h < \xi < x_0 + h$. If $f^{(4)}$ is continuous on $[x_0 - h, x_0 + h]$ it is also bounded, and the approximation is $O(h^2)$.

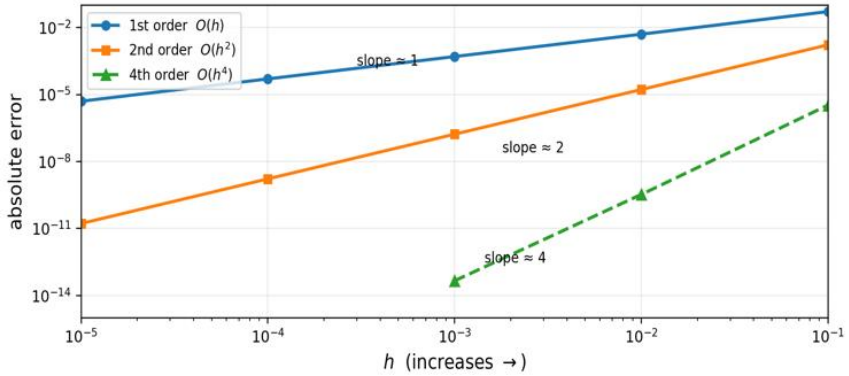
NOTE: It is particularly important to pay attention to round-off error when approximating derivatives.

Example: Numerical differentiation of $f(x) = e^x$ at $x = 0$

Approximate $f'(0)$ for $f(x) = e^x$ using methods of order 1, 2 and 4.
Record and plot the (absolute) errors for $h = 10^{-k},\ k = 1, \ldots, 5$.

Exact derivative:
$f'(0) = e^0 = 1.$

h	1st order (forward) $\left\lvert \frac{e^h - 1}{h} - 1 \right\rvert$	2nd order (central) $\left\lvert \frac{e^h - e^{-h}}{2h} - 1 \right\rvert$	4th order (five-point) $\left\lvert \frac{-e^{2h} + 8e^h - 8e^{-h} + e^{-2h}}{12h} - 1 \right\rvert$
10^{-1}	5.170e-02	1.670e-03	3.33e-06
10^{-2}	5.017e-03	1.667e-05	3.33e-10
10^{-3}	5.002e-04	1.667e-07	4.54e-14
10^{-4}	5.000e-05	1.667e-09	2.60e-13
10^{-5}	5.000e-06	1.667e-11	3.63e-12



Finite difference formulas

1 First-order (forward) difference

$$f'(0) \approx \frac{f(h) - f(0)}{h} = \frac{e^h - 1}{h} \qquad \text{(order } O(h)\text{)}$$

2 Second-order (central) difference

$$f'(0) \approx \frac{f(h) - f(-h)}{2h} = \frac{e^h - e^{-h}}{2h} \qquad \text{(order } O(h^2)\text{)}$$

4 Fourth-order (five-point) difference

$$f'(0) \approx \frac{-f(2h) + 8f(h) - 8f(-h) + f(-2h)}{12h}$$
$$= \frac{-e^{2h} + 8e^h - 8e^{-h} + e^{-2h}}{12h} \qquad \text{(order } O(h^4)\text{)}$$

Observations

- The forward difference is first-order: error $\propto h$.
- The central difference is second-order: error $\propto h^2$.
- The five-point difference is fourth-order: error $\propto h^4$.
- On the log-log plot, the slopes are approximately 1, 2 and 4, confirming the expected orders of accuracy.

Difference formulas

Notation: $f_j = f(x_0 + jh),\ j = 0, \pm 1, \pm 2, \ldots$

First derivative

Scaled derivative	points	type	order	formula
$hf'(x_0)$	2	forward	1	$f_1 - f_0$
$hf'(x_0)$	2	backward	1	$f_0 - f_{-1}$
$2hf'(x_0)$	3	centered	2	$f_1 - f_{-1}$
$2hf'(x_0)$	3	forward	2	$-3f_0 + 4f_1 - f_2$
$12hf'(x_0)$	5	centered	4	$f_{-2} - 8f_{-1} + 8f_1 - f_2$

Second derivative

Scaled derivative	points	type	order	formula
$h^2 f''(x_0)$	3	centered	2	$f_{-1} - 2f_0 + f_1$
$12h^2 f''(x_0)$	5	centered	4	$-f_{-2} + 16f_{-1} - 30f_0 + 16f_1 - f_2$

To obtain the derivative, divide the stencil formula by the scaling factor shown in the first column.

Derivation of the centered difference formula for the second derivative

1 Taylor expansion at $x_0 + h$

$$f(x_0 + h) = f(x_0) + f'(x_0)h + \frac{1}{2} f''(x_0)h^2 + \frac{1}{6} f'''(x_0)h^3 + \frac{1}{24} f^{(4)}(\xi_1)h^4.$$

2 Taylor expansion at $x_0 - h$

$$f(x_0 - h) = f(x_0) - f'(x_0)h + \frac{1}{2} f''(x_0)h^2 - \frac{1}{6} f'''(x_0)h^3 + \frac{1}{24} f^{(4)}(\xi_{-1})h^4.$$

where $x_0 - h < \xi_{-1} < x_0 < \xi_1 < x_0 + h.$

3 Add the two expansions

$$f(x_0 + h) + f(x_0 - h) = 2f(x_0) + f''(x_0)h^2 + \frac{1}{24} \left[f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1}) \right] h^4.$$

4 Rearrange to obtain the formula

$$f''(x_0) = \frac{1}{h^2} \left[f(x_0 - h) - 2f(x_0) + f(x_0 + h) \right] - \frac{h^2}{24} \left[f^{(4)}(\xi_1) + f^{(4)}(\xi_{-1}) \right].$$

Final formula

$$f''(x_0) = \frac{f(x_0 - h) - 2f(x_0) + f(x_0 + h)}{h^2} - \frac{h^2}{12} f^{(4)}(\xi)$$

Takeaway

Thus, the centered difference approximation for $f''(x_0)$ has truncation error of order $O(h^2)$.

Richardson method

Example of method derivation

1 Goal

Goal: Derive a centred, **4th** order formula for the **2nd** derivative.

2 Centred 3-point formula at two step sizes

- Write the centred 3-point formula for $f''(x_0)$ once for h and once for $2h$,

$$\begin{aligned} f''(x_0) &= \frac{1}{h^2}\left(f_{-1} - 2f_0 + f_1\right) - \frac{h^2}{12} f^{(iv)}(x_0) + \mathcal{O}(h^4) \\ &= \frac{1}{(2h)^2}\left(f_{-2} - 2f_0 + f_2\right) - \frac{(2h)^2}{12} f^{(iv)}(x_0) + \mathcal{O}(h^4). \end{aligned}$$

3 Eliminate the leading error term

- The leading term of the truncation error in the second line is 4 times larger than in the first. So multiply the first line by 4 and subtract the **2nd** line:

$$3f''(x_0) = \frac{4}{h^2}\left(f_{-1} - 2f_0 + f_1\right) - \frac{1}{(2h)^2}\left(f_{-2} - 2f_0 + f_2\right) + \mathcal{O}(h^4).$$

4 Rearrangement: 4th order formula

- Rearrange:

$$f''(x_0) = \frac{1}{12h^2}\left(-f_{-2} + 16f_{-1} - 30f_0 + 16f_1 - f_2\right) + \mathcal{O}(h^4).$$

Taylor's theorem can be used to show that centered-difference formula in Eq. (4.5) to approximate $f'(x_0)$ can be expressed with an error formula:

$$f'(x_0) = \frac{1}{2h}[f(x_0 + h) - f(x_0 - h)] - \frac{h^2}{6}f'''(x_0) - \frac{h^4}{120}f^{(5)}(x_0) - \cdots .$$

Find approximations of order $O(h^2)$, $O(h^4)$, and $O(h^6)$ for $f'(2.0)$ when $f(x) = xe^x$ and $h = 0.2$.

Truncation error of the centred difference formula

Result (from Taylor's theorem)

The centred-difference formula for the first derivative can be written with an error term as

$$f'(x_0) = \frac{1}{2h}\Big[f(x_0+h) - f(x_0-h)\Big] - \frac{h^2}{6}f^{(3)}(x_0) - \frac{h^4}{120}f^{(5)}(x_0) - \cdots$$

Example

Find approximations of order $O(h^2)$, $O(h^4)$, and $O(h^6)$ for $f'(2.0)$ when

$$f(x) = xe^x$$

and $h = 0.2$.

Richardson extrapolation for $N_2(h)$ (second derivative)

1 Error expansion

$$M(h) = N_2(h) = N_2\left(\frac{h}{2}\right) + K_1\frac{h^2}{4} + K_2\frac{h^4}{16} + K_3\frac{h^6}{64} + \cdots .$$

2 Eliminate the h^2 term

Subtract 4 times the equation with $h/2$ from the one with h:

$$\begin{aligned} 3M(h) &= 4N_2\left(\frac{h}{2}\right) - N_2(h) + K_2\left(\frac{h^4}{4} - h^4\right) + K_3\left(\frac{h^6}{16} - h^6\right) + \cdots \\ &= 4N_2\left(\frac{h}{2}\right) - N_2(h) - \frac{3}{4}K_2h^4 - \frac{15}{16}K_3h^6 + \cdots . \end{aligned}$$

3 Divide by 3 ($O(h^4)$ formula)

$$M(h) = \frac{1}{3}\left[4N_2\left(\frac{h}{2}\right) - N_2(h)\right] + K_2\frac{h^4}{4} - K_3\frac{5h^6}{16} + \cdots . \tag{4.1}$$

4 Eliminate the h^4 term

Replace h in (4.1) by $h/2$:

$$M(h) = N_2\left(\frac{h}{2}\right) - K_2\frac{h^4}{64} - K_3\frac{5h^6}{1024} + \cdots .$$

5 Subtract 16 times (eliminate h^4)

Subtract 16 times the equation with $h/2$ from the one with h:

$$15M(h) = 16N_2\left(\frac{h}{2}\right) - N_2(h) + K_3\frac{15h^6}{64} + \cdots .$$

6 Divide by 15 ($O(h^6)$ formula)

Finally, we obtain the $O(h^6)$ approximation

$$M(h) = \frac{1}{15}\left[16N_2\left(\frac{h}{2}\right) - N_2(h)\right] + K_3\frac{h^6}{64} + \cdots .$$

General Richardson extrapolation

For a sequence of approximations with even-order error expansion

$$N_j(h) = N + K_1h^2 + K_2h^4 + \cdots + K_jh^{2j} + \cdots ,$$

the Richardson extrapolation formula is

$$N_{j+1}(h) = N_j\left(\frac{h}{2}\right) + \frac{N_j(h/2) - N_j(h)}{4^{j-1} - 1}.$$

Resulting formulas for $N_2(h)$

$O(h^2)$	$N_2(h) = \dfrac{f(x_0+h) - 2f(x_0) + f(x_0-h)}{h^2}$
$O(h^4)$	$N_3(h) = \dfrac{4N_2(h/2) - N_2(h)}{3}$
$O(h^6)$	$N_4(h) = \dfrac{16N_2(h/2) - N_2(h)}{15}$

Richardson extrapolation tableau

$$M = N_1(h) + K_1 h^2 + K_2 h^4 + K_3 h^6 + \cdots .$$

Table 4.6

$O(h^2)$	$O(h^4)$	$O(h^6)$	$O(h^8)$
1: $N_1(h)$			
2: $N_1\left(\frac{h}{2}\right)$	**3:** $N_2(h)$		
4: $N_1\left(\frac{h}{4}\right)$	**5:** $N_2\left(\frac{h}{2}\right)$	**6:** $N_3(h)$	
7: $N_1\left(\frac{h}{8}\right)$	**8:** $N_2\left(\frac{h}{4}\right)$	**9:** $N_3\left(\frac{h}{2}\right)$	**10:** $N_4(h)$

Takeaway: | Each step halving h and moving diagonally right increases the order of accuracy.

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